

Chapters 26

Signal Coordination for Arterials and Networks: Undersaturated Conditions

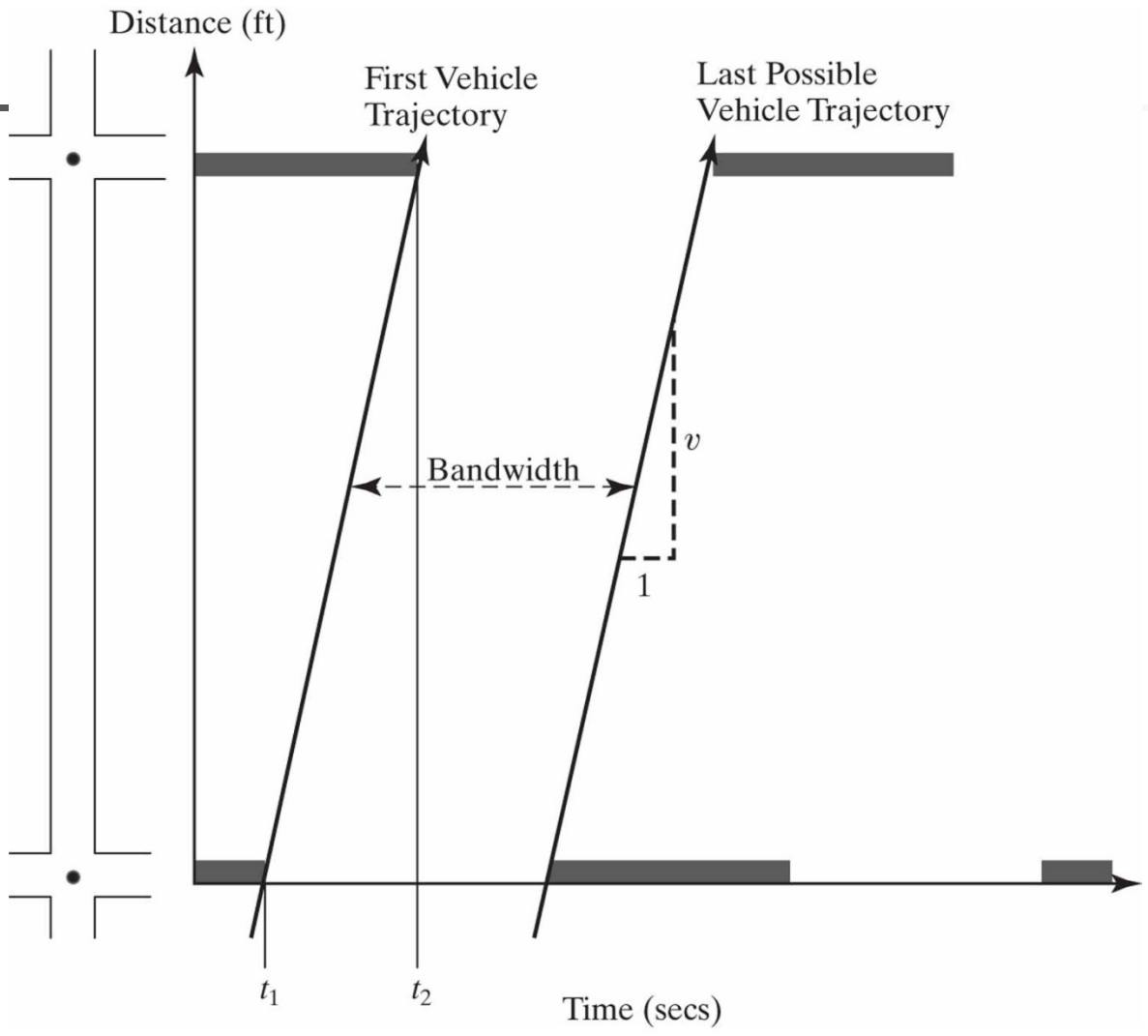
Learning Objectives

- Determine progression offset, bandwidth, bandwidth efficiency, and bandwidth capacity.
- Develop progression for one-way streets and two-way streets.

Introduction

- When signals are close enough together, it is beneficial to coordinate their starting green times so that vehicles may move efficiently through the corridor.
- To make progression work, all intersections in the system must have the same cycle length or multiples of the minimum cycle length (long enough to provide sufficient capacity a the busiest intersection).

Illustration of signal progression via time-space diagram



$t_2 - t_1 \Rightarrow$ offset

Ideal offset

- Ideal offset is defined as time needed for the first vehicle in the platoon to reach the downstream signal as it turns green.

$$t_{ideal} = \frac{L}{S}$$

where:

t_{ideal} = ideal offset, sec

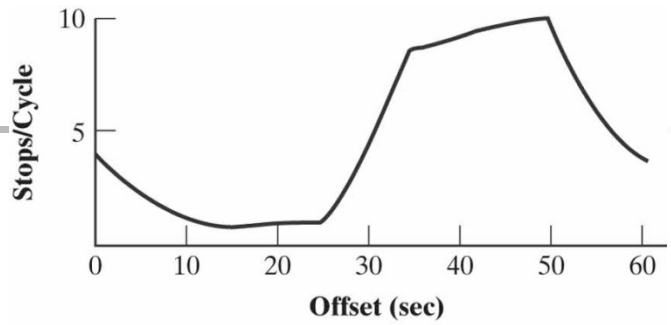
L = distance between signalized intersections, ft

S = vehicle speed, ft/s

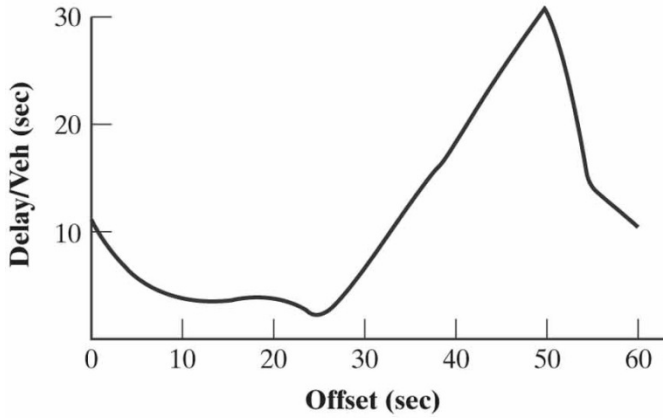
- If vehicles were stopped and had to accelerate after some initial startup delay, 2 to 4 seconds should be added to the first intersection.

Effect of offset on stops and delay

Note that an offset of 25+10 = 35 sec causes more harm than an offset of 25–10 = 15 sec.



(a) Stops



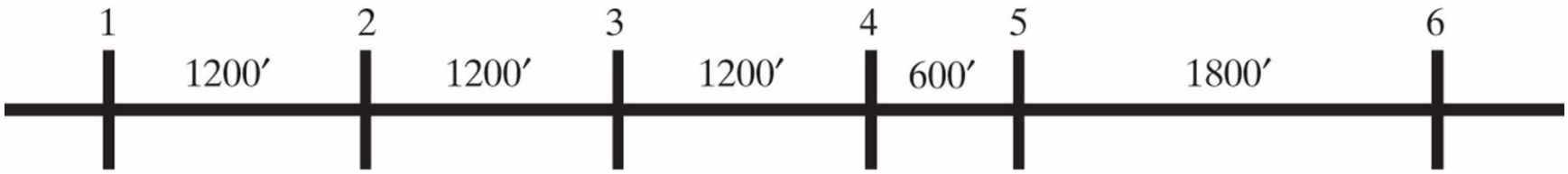
(b) Delay

600 ft block
600 vph in two lanes
all through traffic
free speed 24 mph
50-50 split
60 sec. cycle length

Copyright © 2011 Pearson Education, Inc. publishing as Prentice Hall

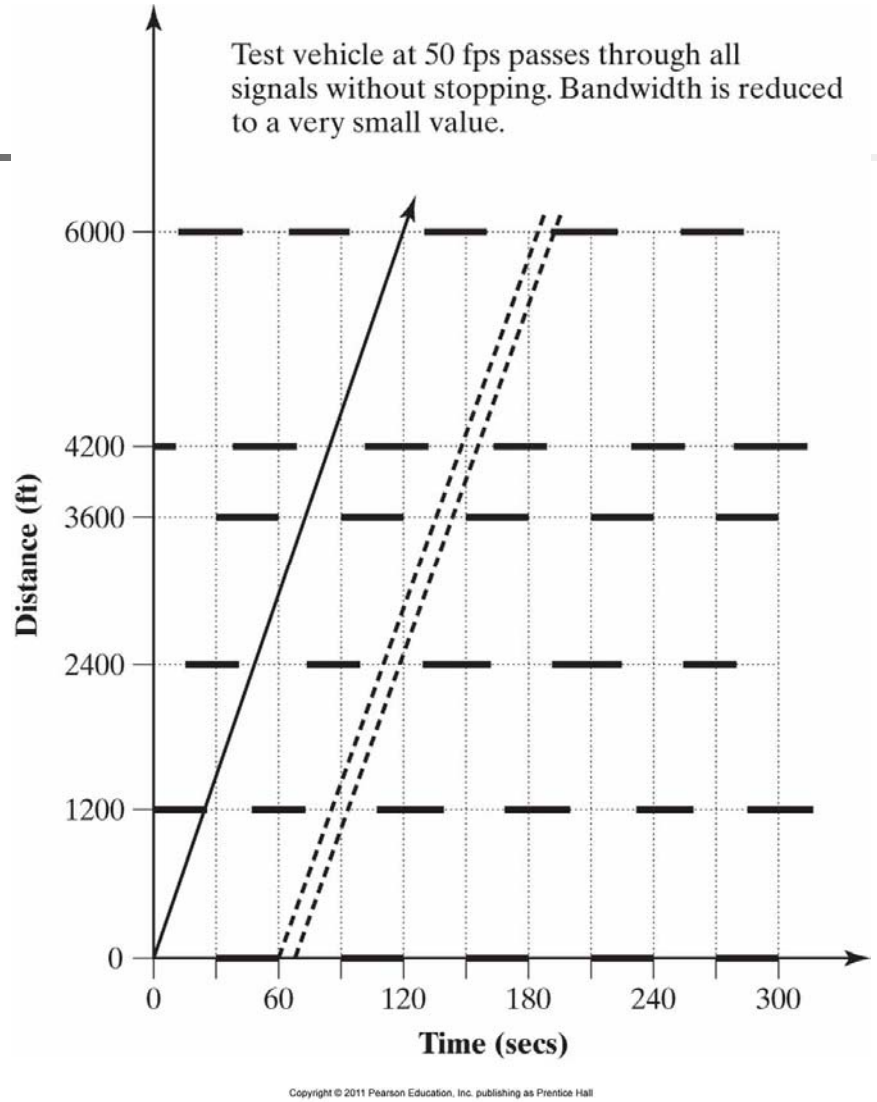
Progression on a one-way street

Consider the following one-way arterial. No vehicles are queued at the signals and the platoon speed is 60 ft/s. The cycle length is 60 seconds and the effective green time at each intersection is 50% of the cycle length. Determine the ideal offsets and Illustrate the progression via the time-space diagram.



Copyright © 2011 Pearson Education, Inc. publishing as Prentice Hall

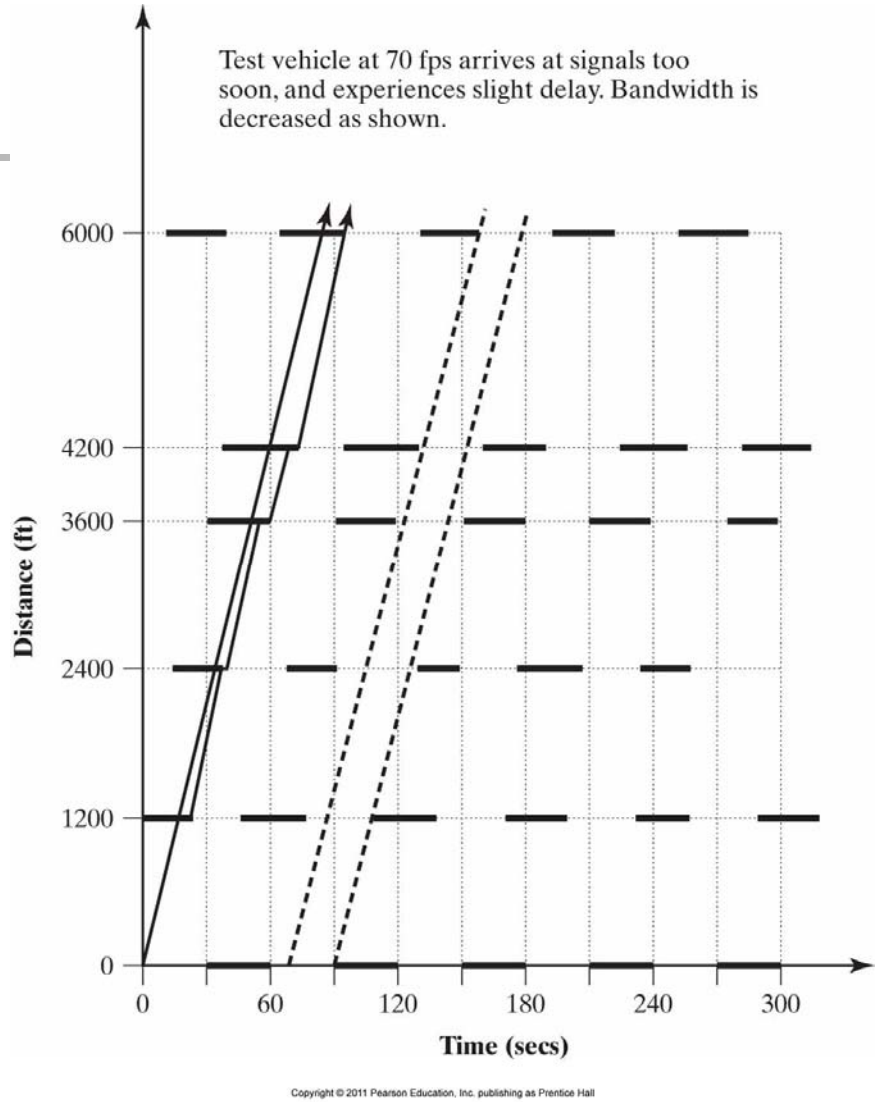
Effect of a 50 ft/s platoon speed on progression



Copyright © 2011 Pearson Education, Inc. publishing as Prentice Hall

Effect of a 70 ft/s platoon speed on progression

Note that the effect on bandwidth is not as severe when platoon speed was at 50 ft/s.



Copyright © 2011 Pearson Education, Inc. publishing as Prentice Hall

Bandwidth efficiency

- The efficiency of bandwidth is defined as the ratio of the bandwidth to the cycle length, expressed as a percentage:

$$EFF_{BW} = \left(\frac{BW}{C} \right) * 100\%$$

where:

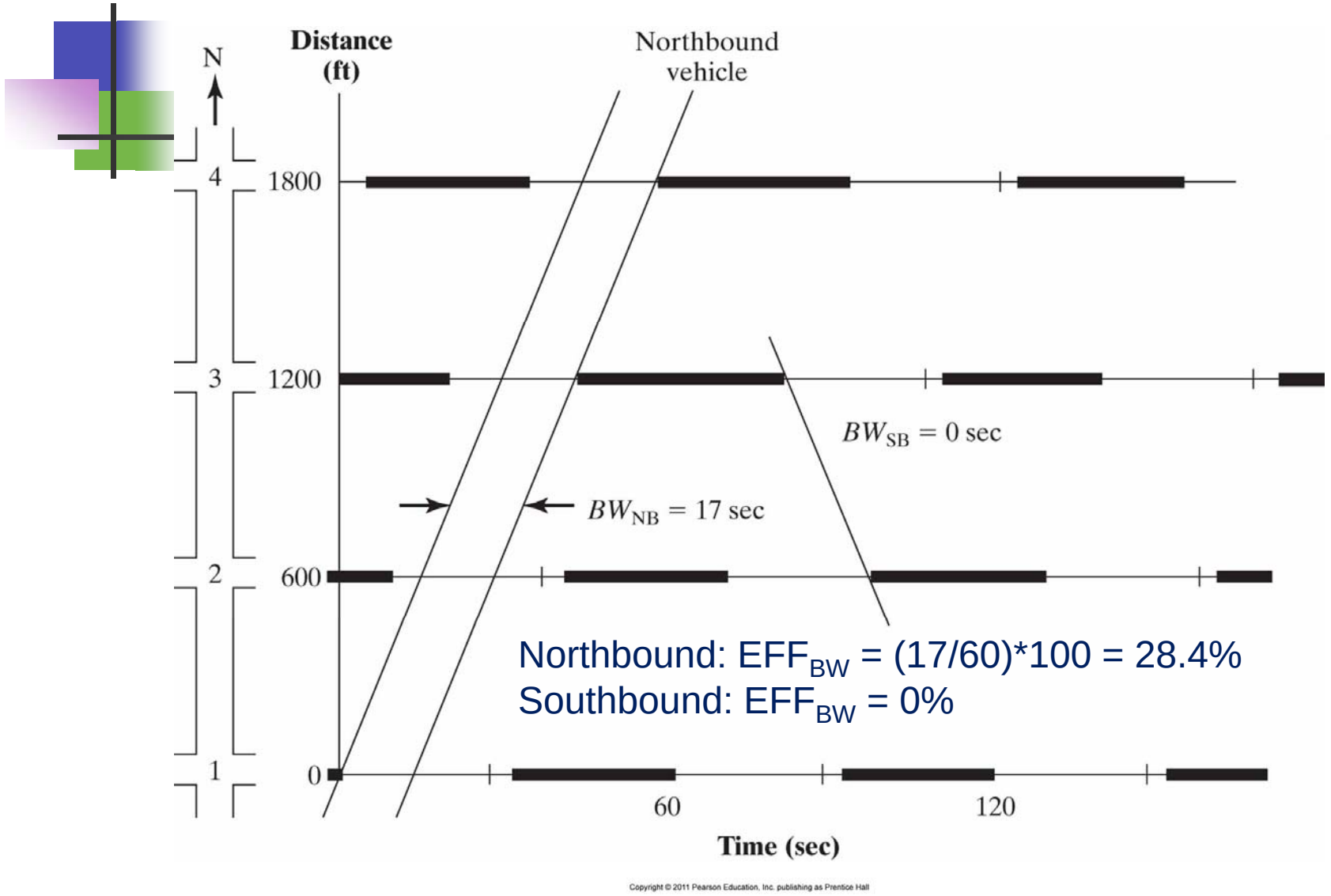
EFF_{BW} = bandwidth efficiency (%)

BW = bandwidth (s)

C = cycle length (s)

- Bandwidth efficiency of 40% to 55% is considered good.

Example of low bandwidth coordinated system



Northbound: EFF_{BW} = (17/60)*100 = 28.4%

Southbound: EFF_{BW} = 0%

Bandwidth capacity

- Bandwidth capacity is defined as the number of vehicles that can pass through the system without stopping.

$$C_{BW} = \frac{3600 * BW * NL}{C * h}$$

where:

C_{BW} = bandwidth capacity, veh/h

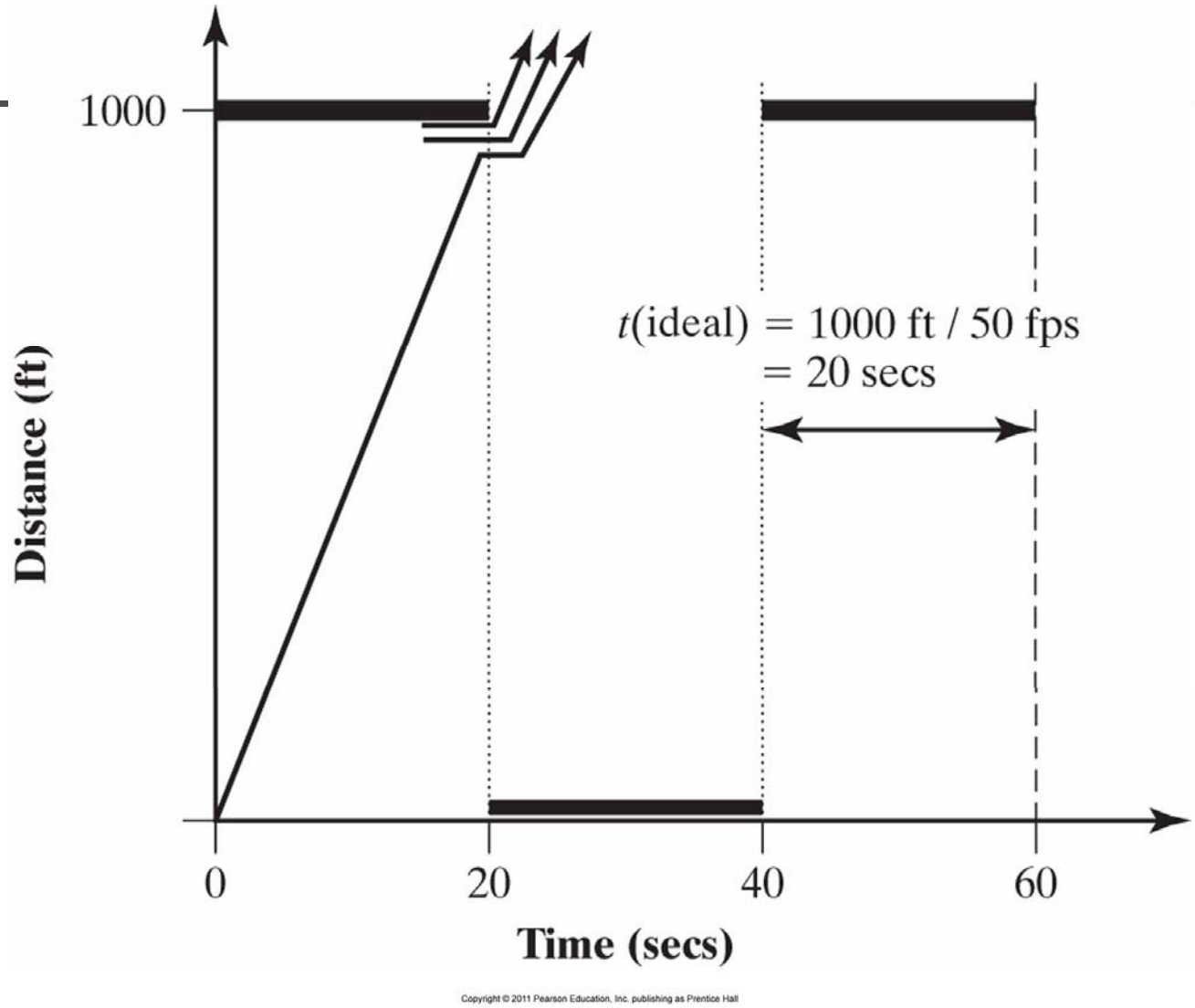
BW = bandwidth, s

NL = number of through lanes in the indicated direction

C = cycle length, s

H = saturation headway, s

Effect of queued vehicles at a signal



Copyright © 2011 Pearson Education, Inc. publishing as Prentice Hall

Adjusted ideal offset

$$t_{adj} = \frac{L}{S} - \left(Qh + l_1\right)$$

where:

t_{adj} = adjusted ideal offset, s

L = distance between signals, ft

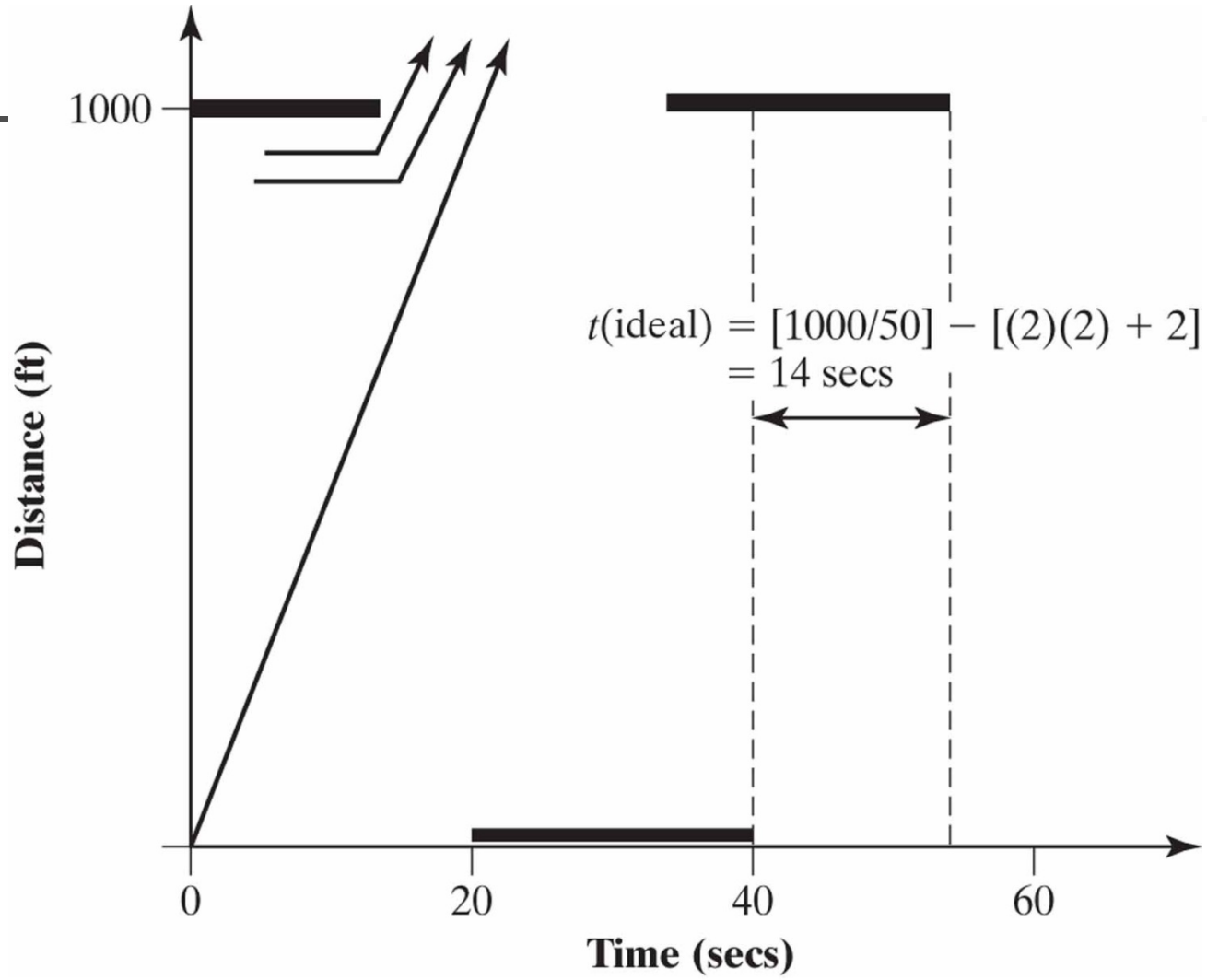
S = speed, ft/s

Q = number of queued vehicles per lane, veh

h = discharge headway of queued vehicles, sec/veh

l_1 = start-up lost time, s

Adjustment in offset to accommodate queued vehicles



Copyright © 2011 Pearson Education, Inc. publishing as Prentice Hall

Effect of queue clearance on progression speed

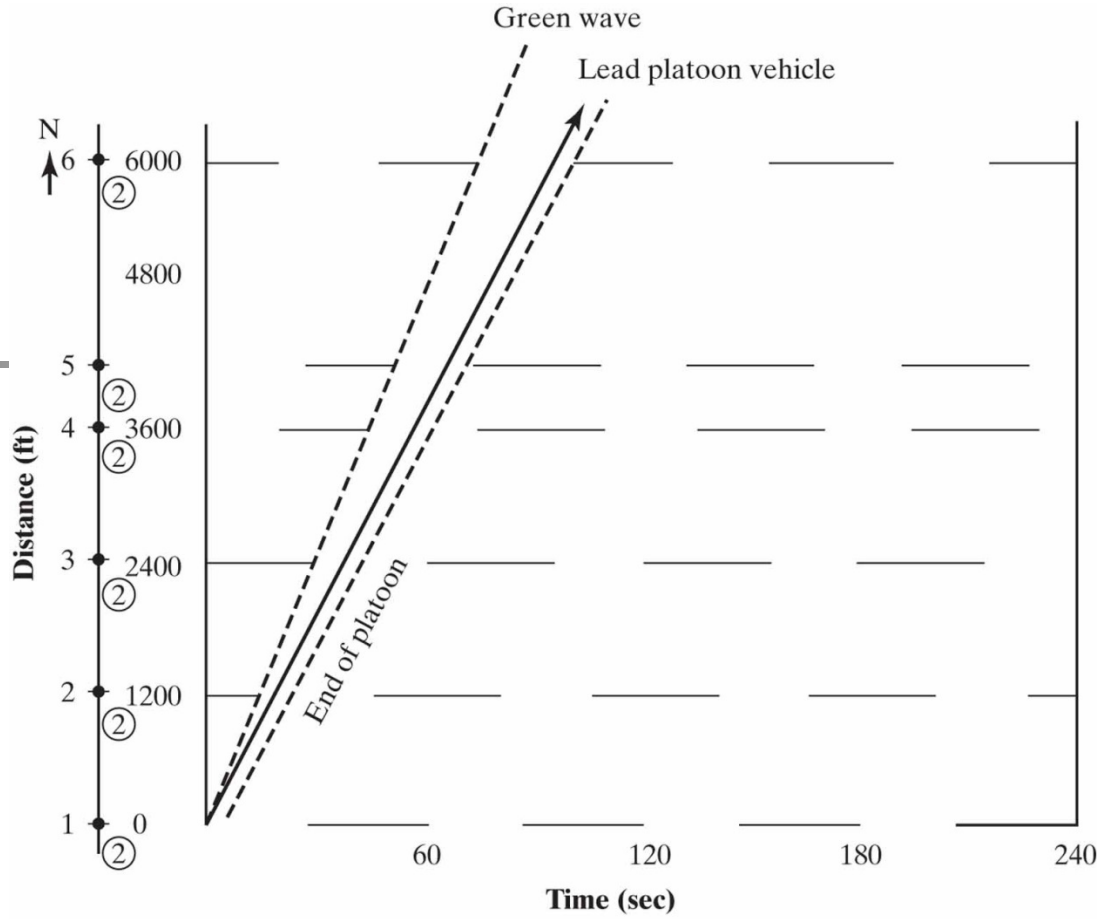
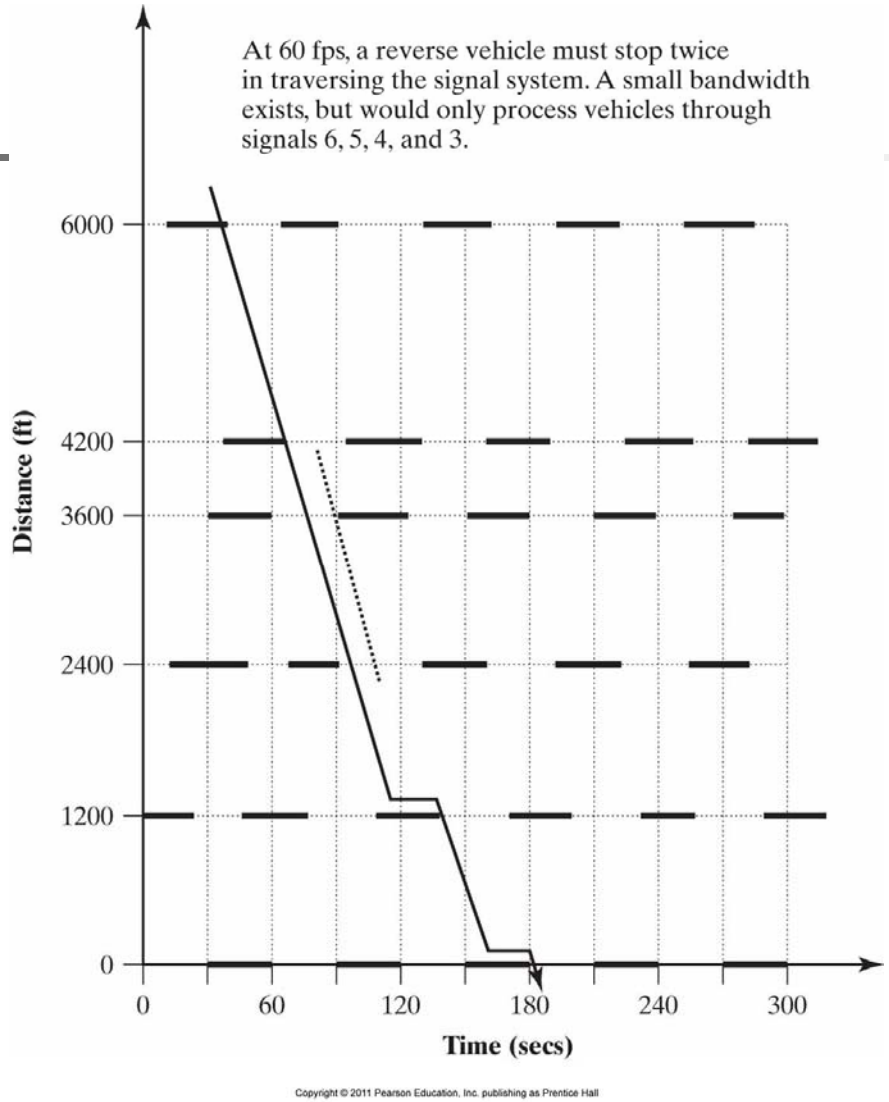


Table 26.2: Progression Speeds in Figure 26-11

Link	Link Offset (s)	Speed of Progression (ft/s)
Signal 1 → 2	(1,200/60) − (4 + 2) = 14	1,200/14 = 85.7
Signal 2 → 3	(1,200/60) − (4) = 16	1,200/16 = 75
Signal 3 → 4	(1,200/60) − (4) = 16	1,200/16 = 75
Signal 4 → 5	(600/60) − (4) = 6	600/6 = 100
Signal 5 → 6	(1,800/60) − (4) = 26	1,800/26 = 69.2

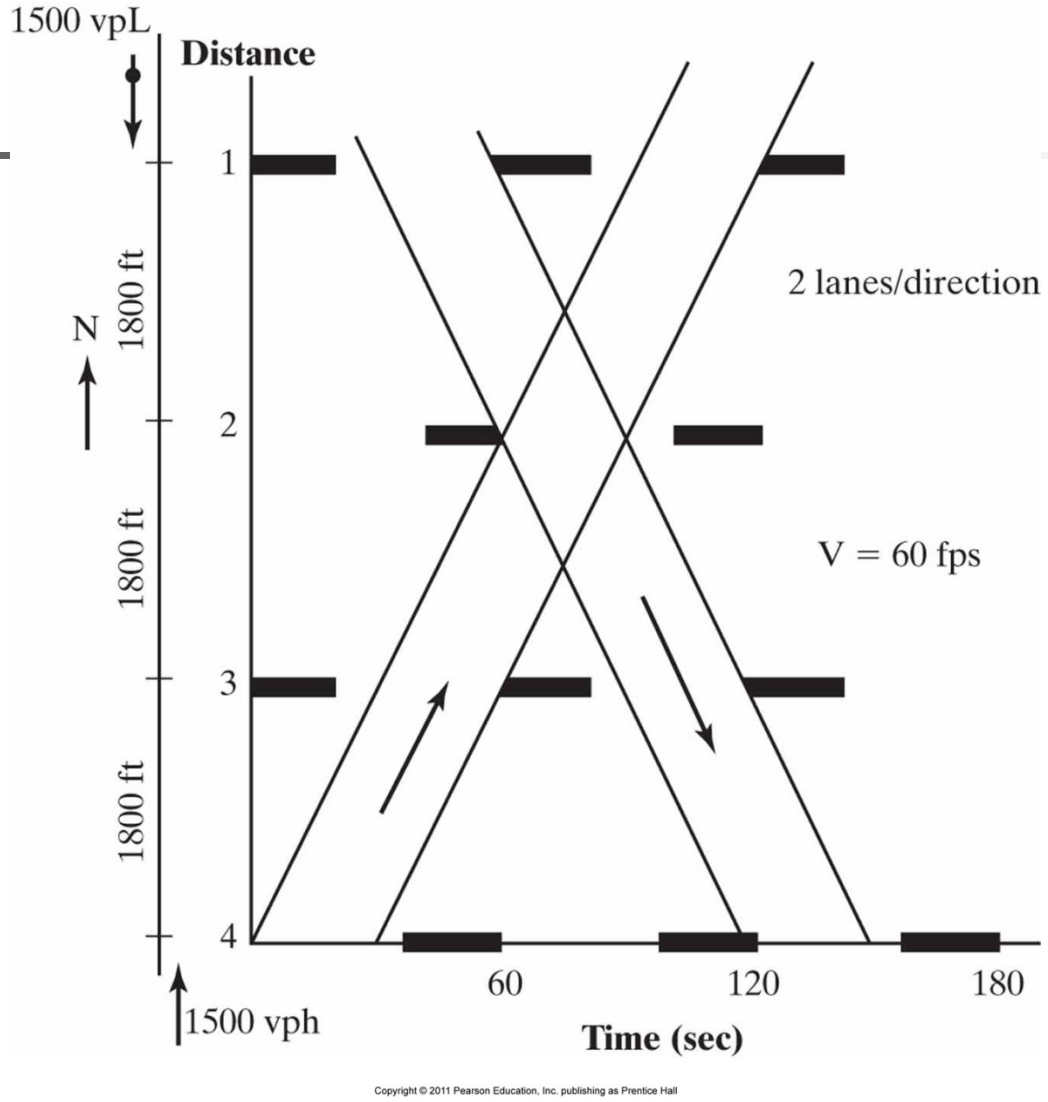
Total Offset = 78 sec

Two-way progression (NB vs. SB)



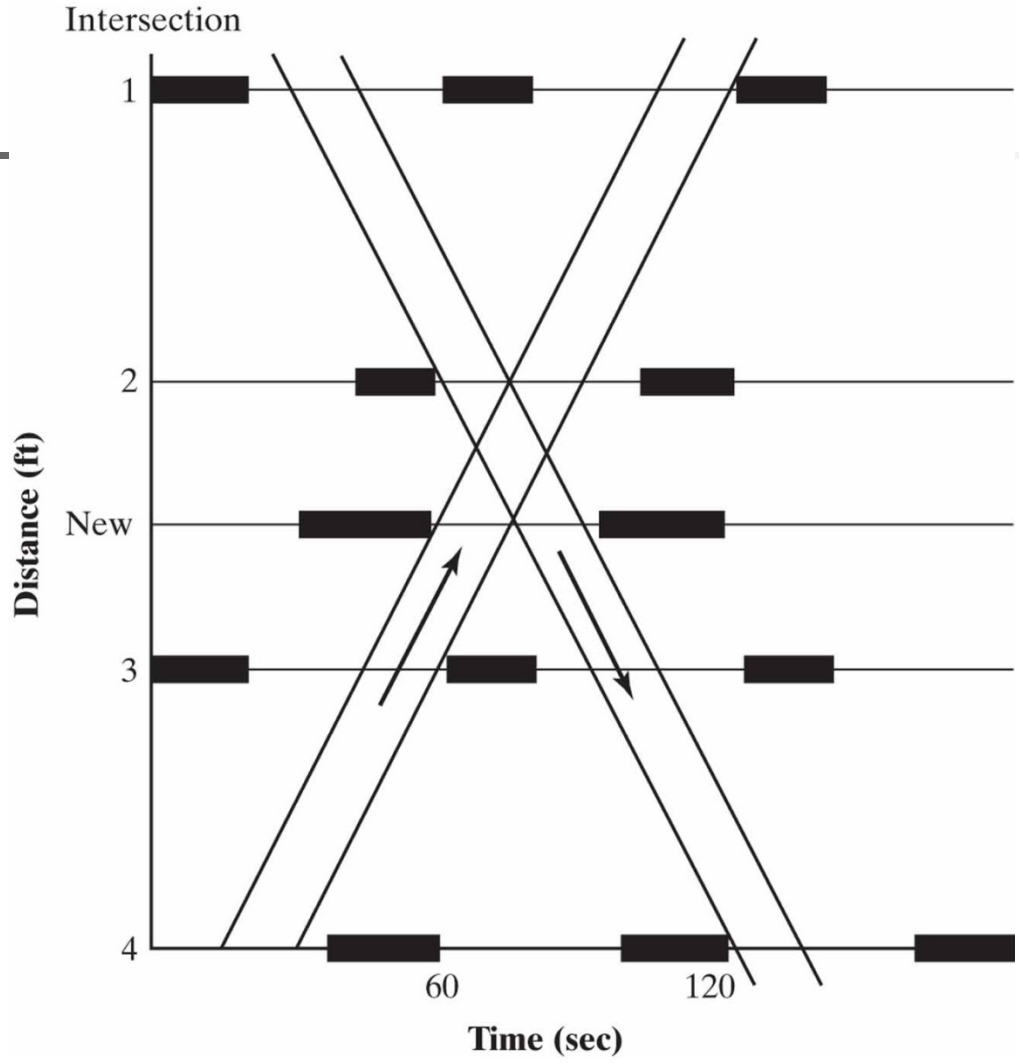
Copyright © 2011 Pearson Education, Inc. publishing as Prentice Hall

Finding solutions (by hand)



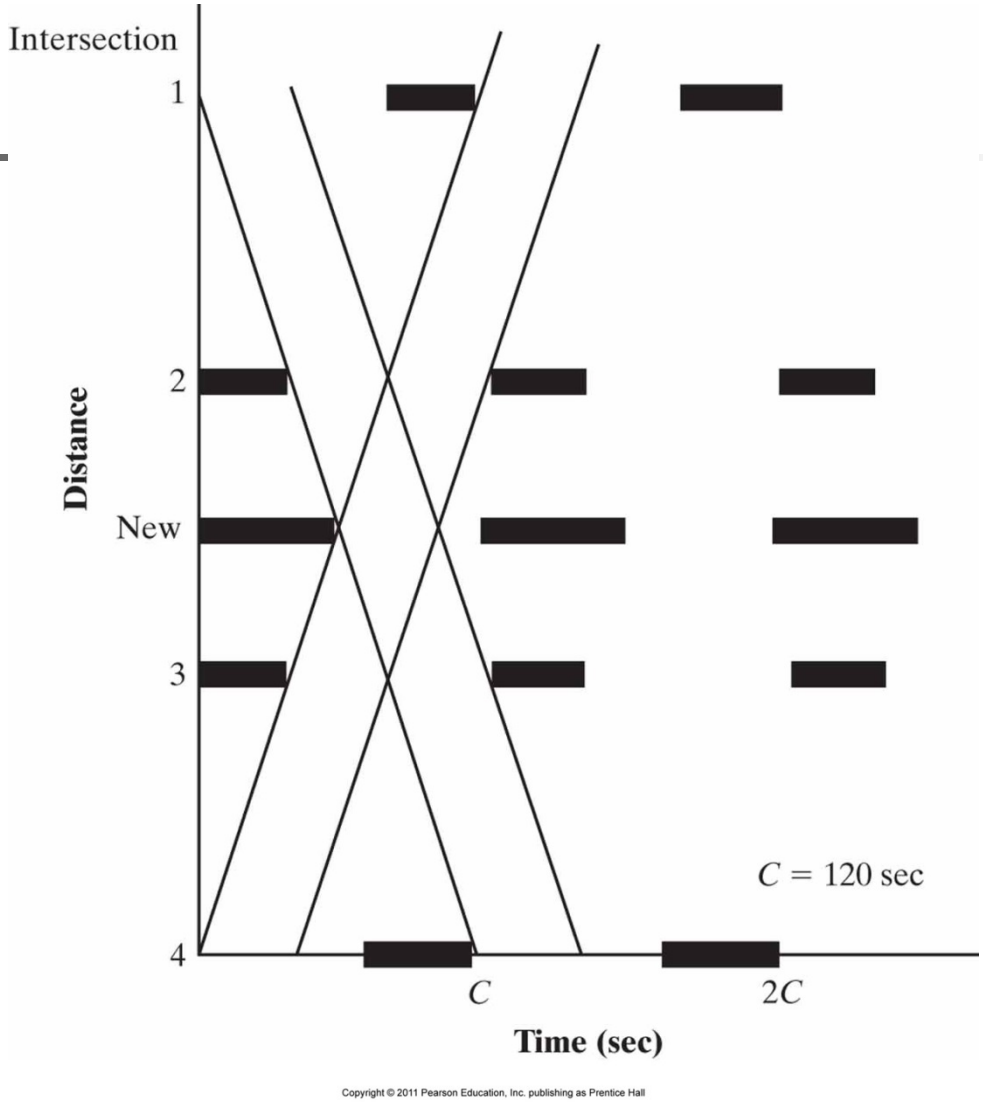
Copyright © 2011 Pearson Education, Inc. publishing as Prentice Hall

Effect of inserting a new signal into the system



Copyright © 2011 Pearson Education, Inc. publishing as Prentice Hall

A Solution with C = 120 Seconds

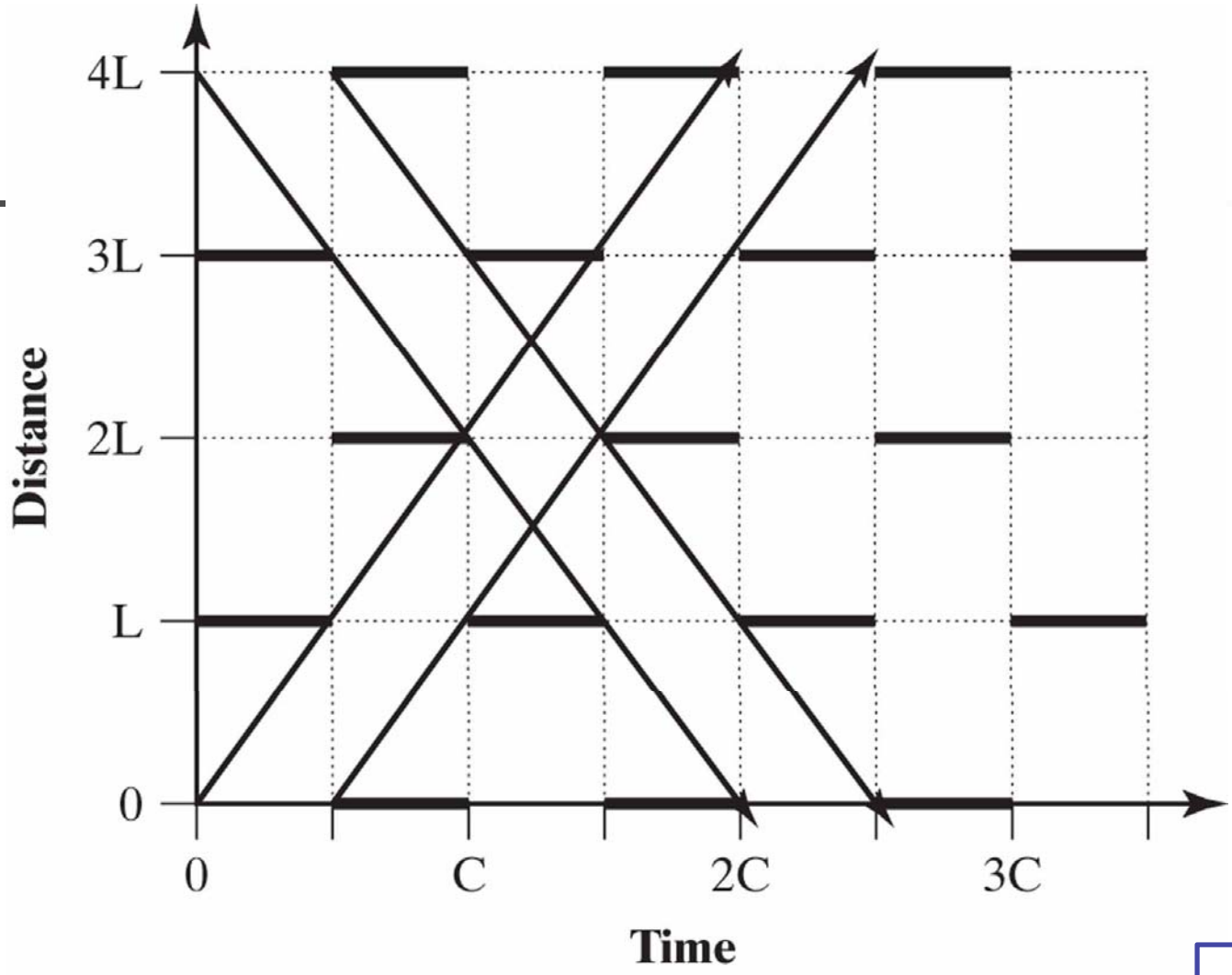


Copyright © 2011 Pearson Education, Inc. publishing as Prentice Hall

Common types of progression

- Single alternate progression
- Double alternate progression
- Simultaneous progression

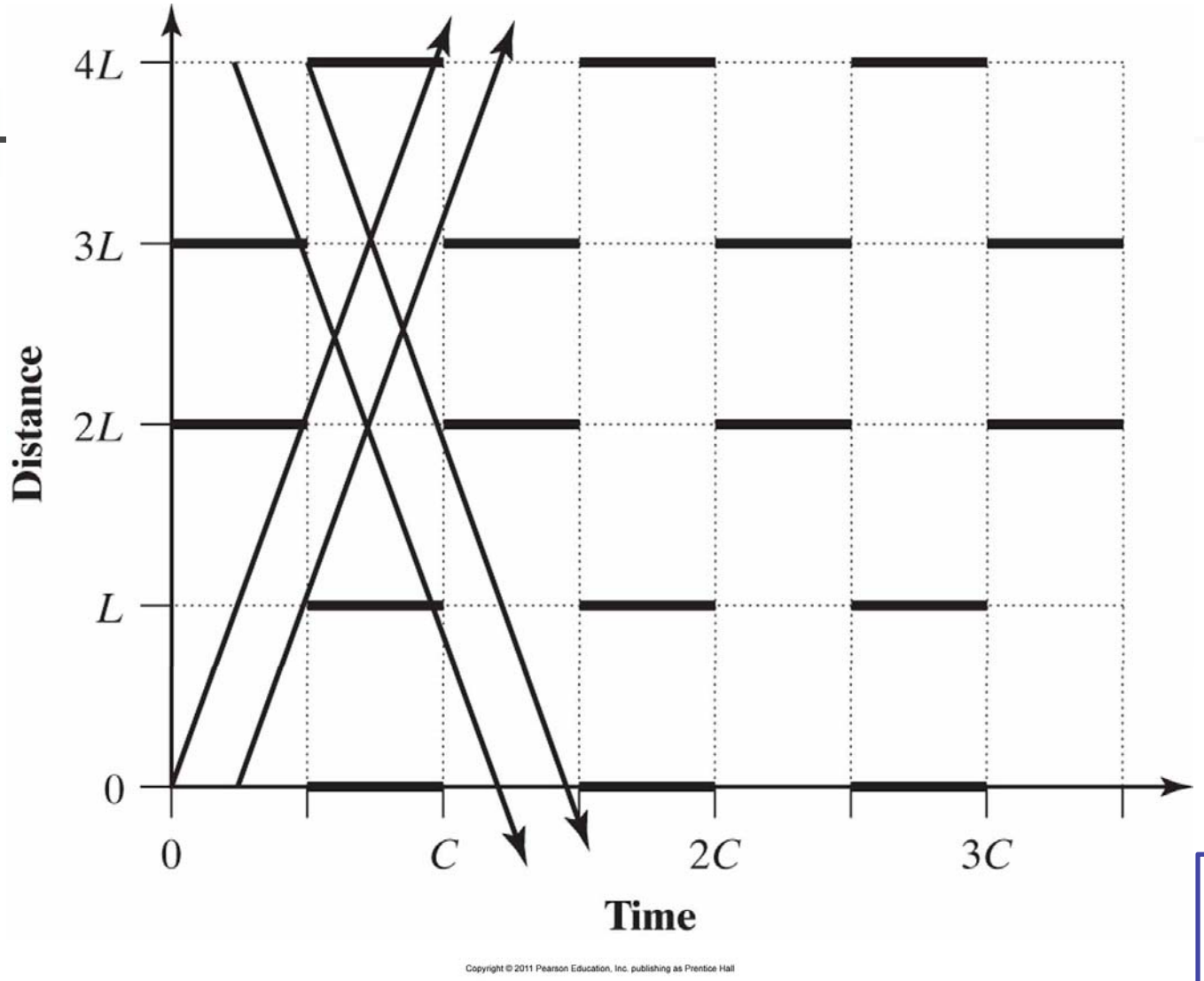
Single alternate progression



Copyright © 2011 Pearson Education, Inc. publishing as Prentice Hall

$$\frac{C}{2} = \frac{L}{S}$$

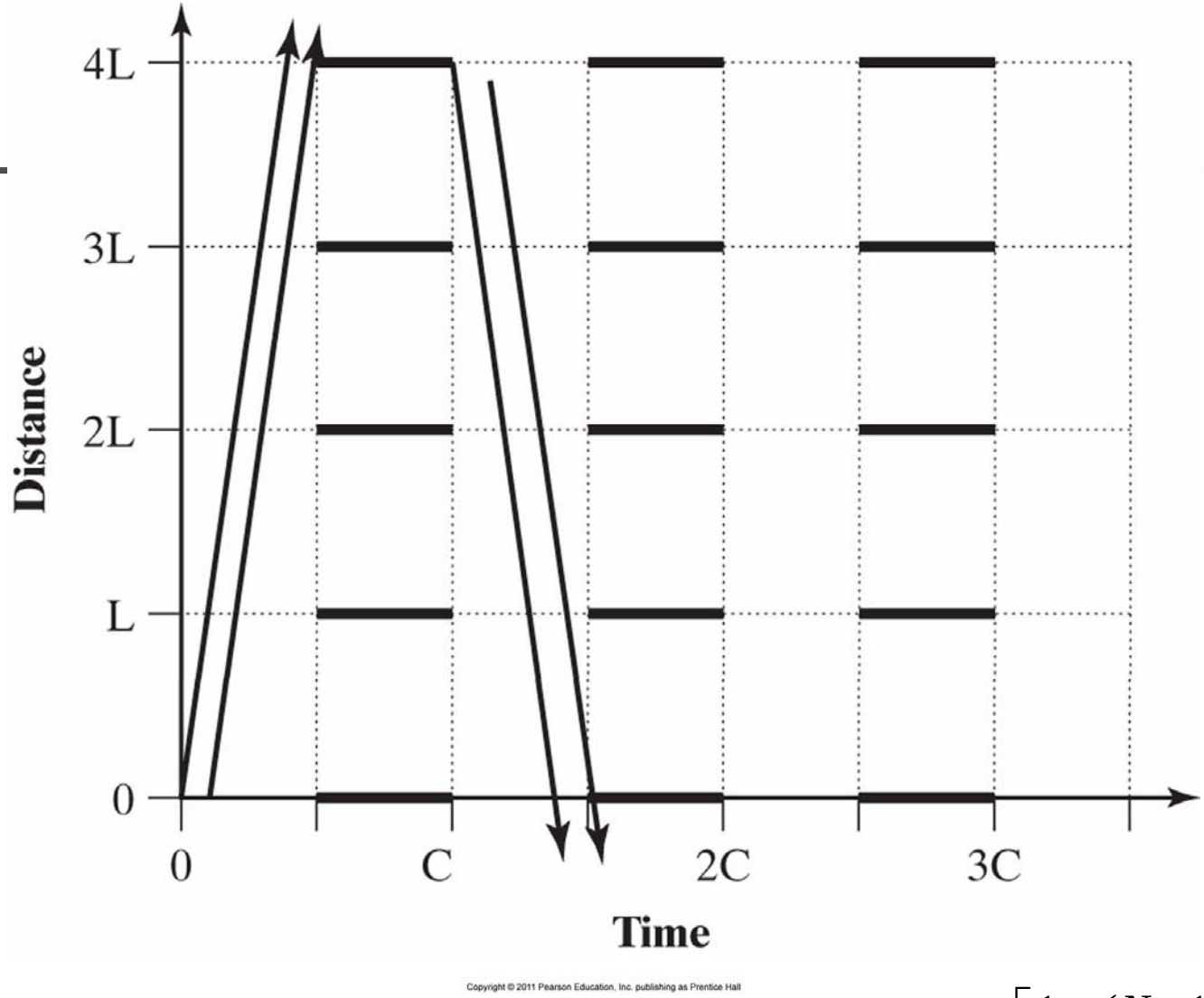
Double alternate progression



Copyright © 2011 Pearson Education, Inc. publishing as Prentice Hall

$$\frac{C}{4} = \frac{L}{S}$$

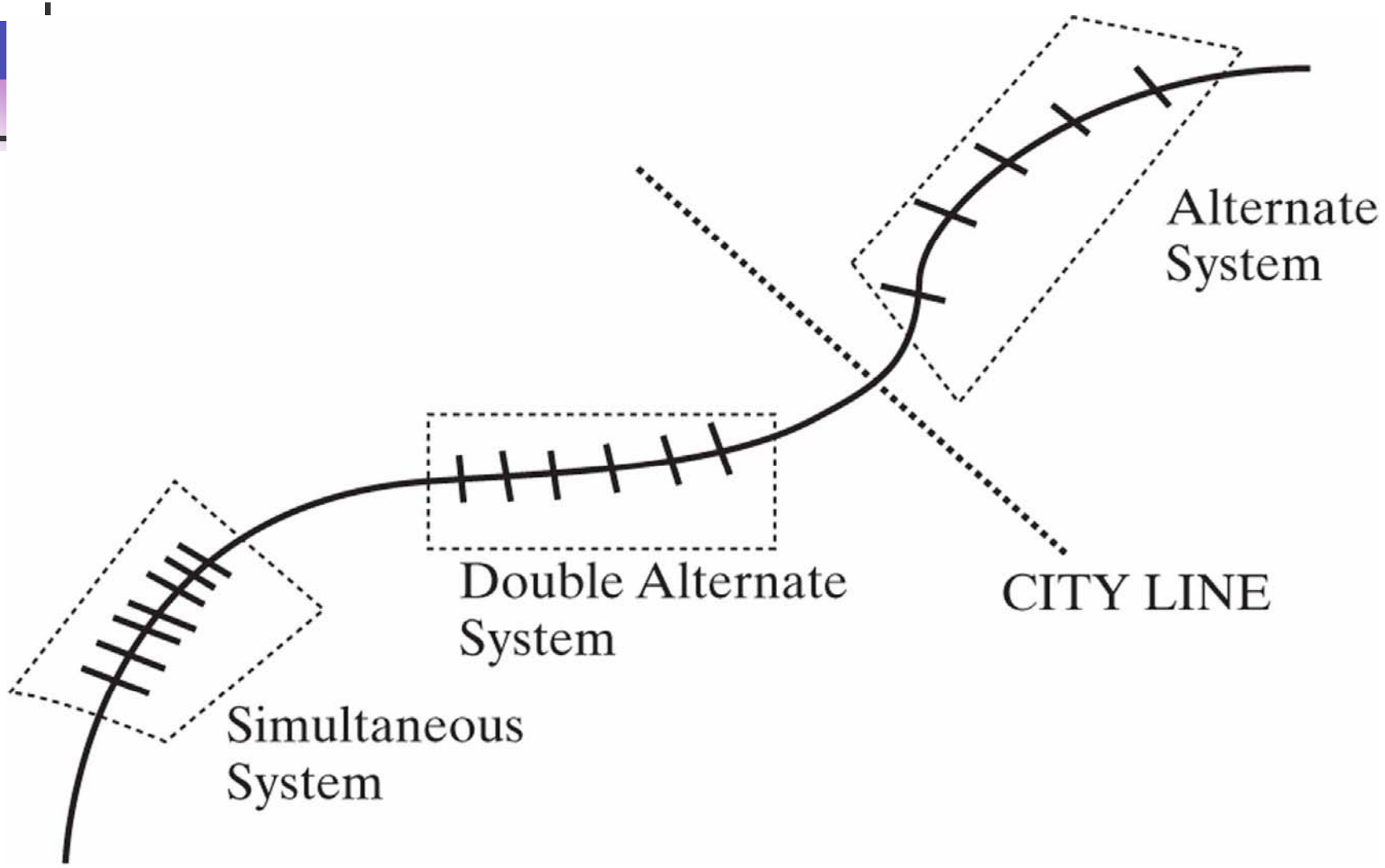
Simultaneous progression



Copyright © 2011 Pearson Education, Inc. publishing as Prentice Hall

$$EFF_{BW} = \left[\frac{1}{2} - \frac{(N-1)\times L}{S\times C}\right]\times 100\%$$

Hypothetical use of several patterns along an arterial



Copyright © 2011 Pearson Education, Inc. publishing as Prentice Hall

Problem 26.2

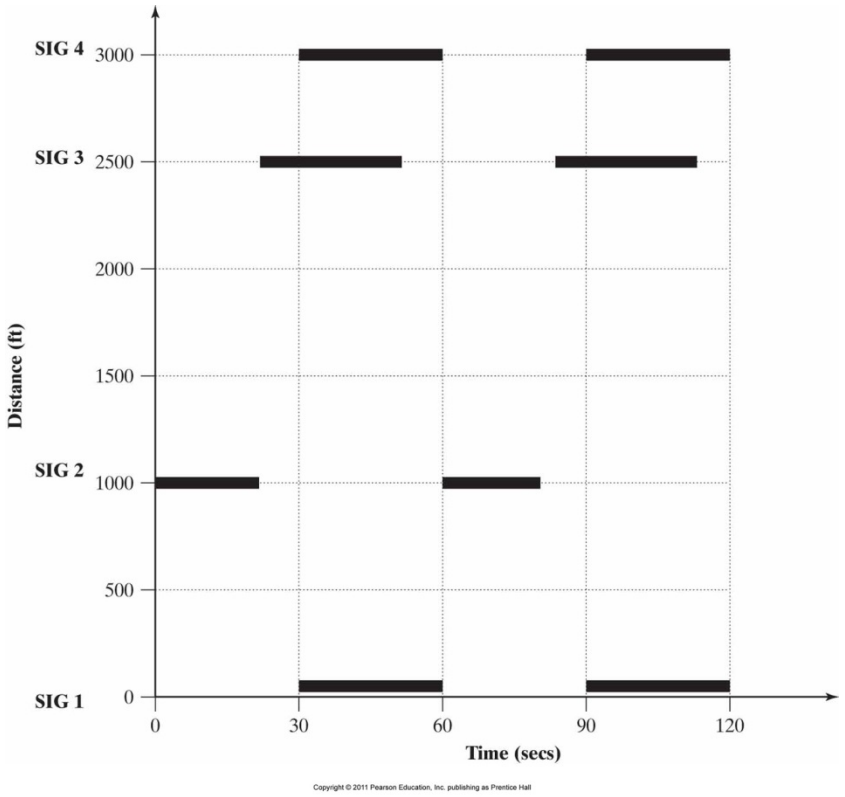
Consider the time-space diagram for this problem.

(a) what is the NB progression speed?

(b) what is the NB bandwidth and bandwidth capacity? Assume a saturation headway of 2 sec/veh.

(c) what is the bandwidth in the SB direction for the same desired speed as the NB progression speed? What is the SB bandwidth capacity?

(d) A new development introduces a major driveway that must be signalized between intersections 2 and 3. It requires 15 seconds of green time out of the 60-second cycle. Assuming that you have complete flexibility as to the exact location of the new driveway, where would you place it? Why?



Copyright © 2011 Pearson Education, Inc. publishing as Prentice Hall

Problem 26.3

A downtown grid has equal block lengths of 750 feet along its primary arterial. It is desired to provide for a progression speed of 30 mi/hr, providing equal service to traffic in both directions along the arterial.

(a) would you suggest a single-alternate or a double-alternate progression scheme? Why?

(b) Assuming your answer to part (a), what cycle length would you suggest?